London is building more data centres. How many direct jobs does a typical facility create? Technical Annex

Repository. Complete code.

Overview

This document presents a step-by-step regression modelling addressing the following main question:

How many people are employed in data centres, and how does this relationship change with DC size or other characteristics?

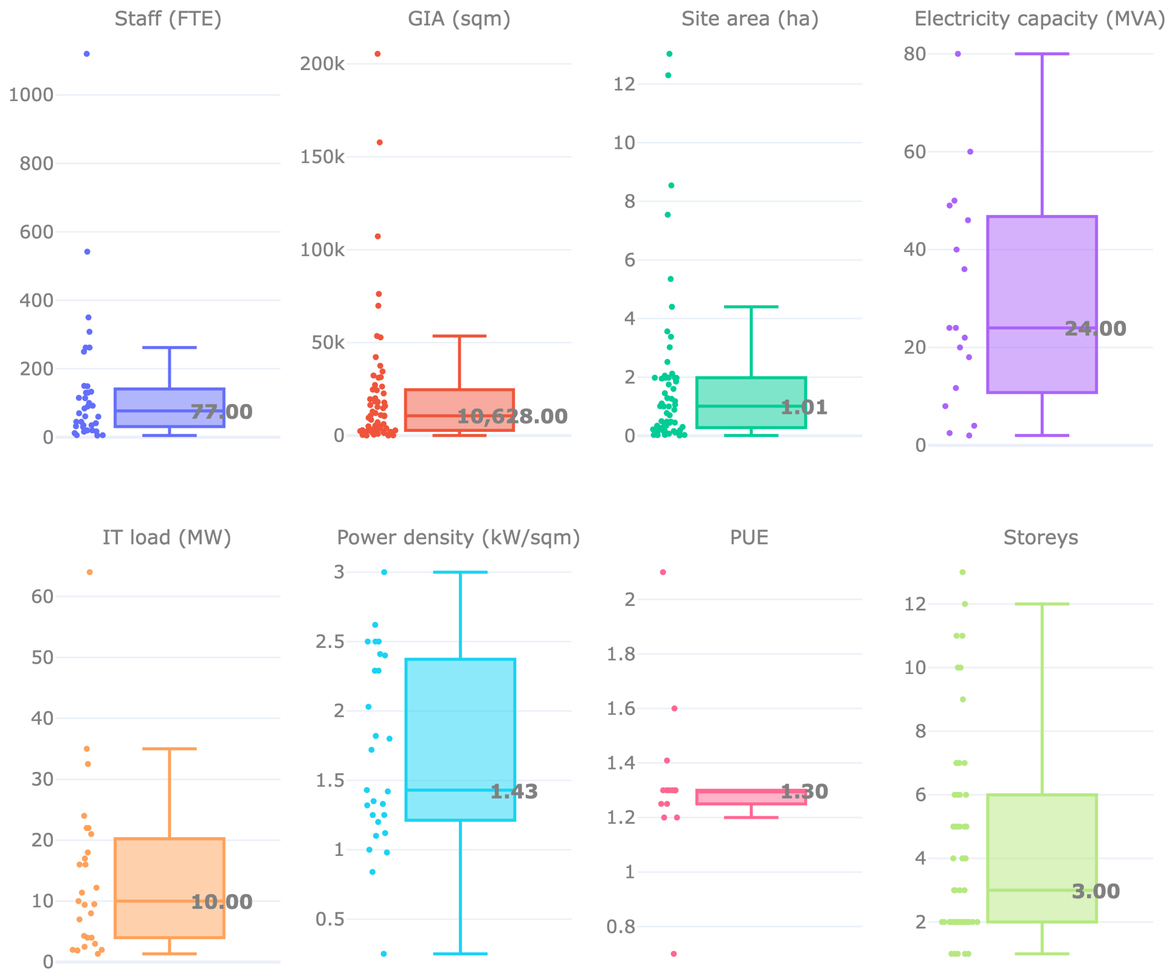
We used the Planning London Datahub API to acquire a sample of DC using the keywords. The sample is restricted to DC in London only. The dataset contains N=67 observations; regression samples are further reduced by complete-case availability. **The main GIA–staff model uses N=36**. Additional specifications use N=35 and N=15.

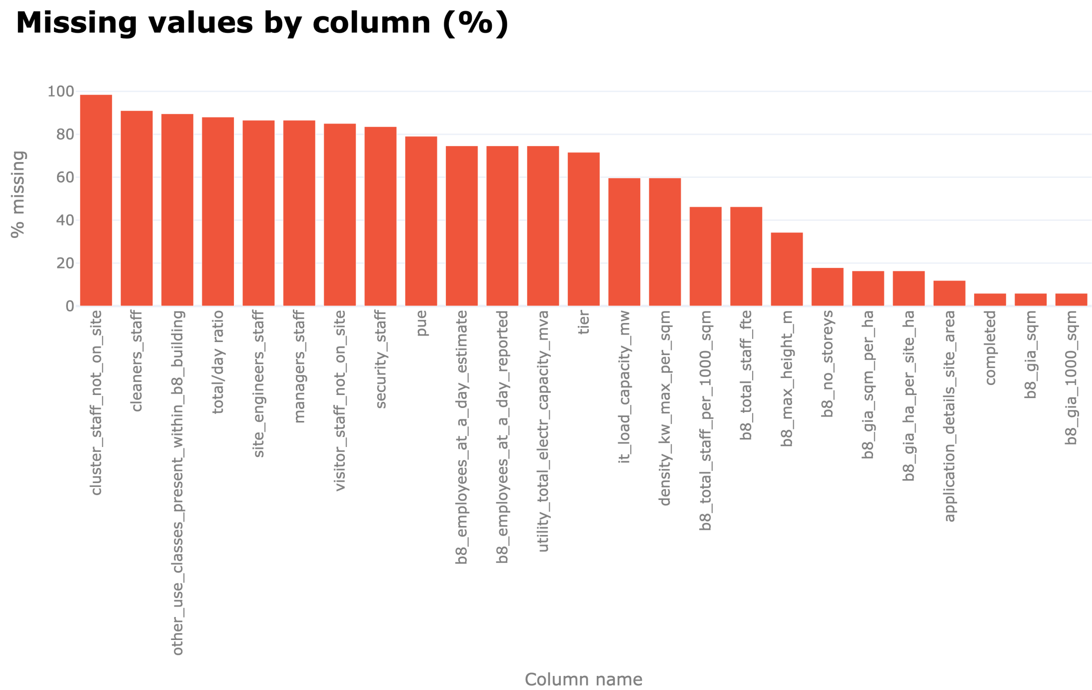
Loading and EDA

The dataset contains key characteristics of each data centre, covering its planning and ownership status, spatial characteristics, technical capacity, and employment profile. A subset of these variables is used in the analysis, including gross internal area (GIA), site area, electricity and IT load capacity, power density, total staff numbers, and the spatial location of each data centre.

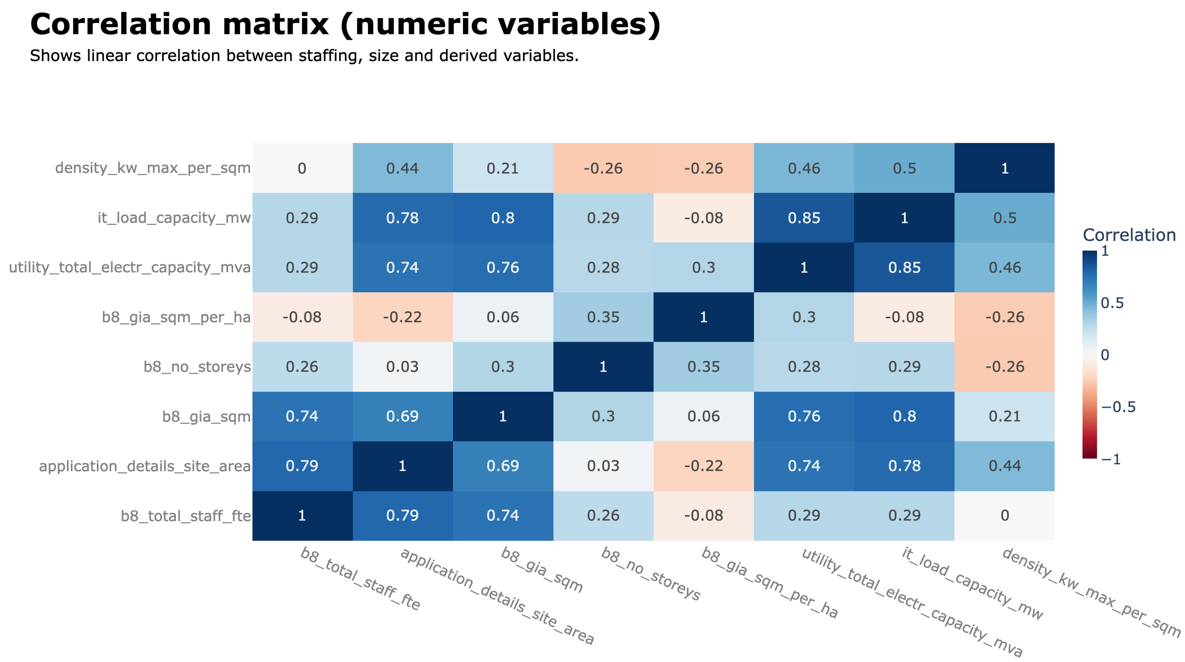
	Staff	GIA	Site_ha	Elec.capacity	IT load	Max density
count	36.0	63.0	59.0	17.0	27.0	27.0
mean	137.1	21792.8	1.8	29.2	14.1	1.7
std	204.8	35800.6	2.7	22.3	13.7	0.7
min	5.0	24.0	0.0	2.0	1.4	0.2
25%	31.5	2812.5	0.3	11.7	4.0	1.2
50%	77.0	10628.0	1.0	24.0	10.0	1.4
75%	137.0	24547.0	2.0	46.0	19.5	2.3
max	1120.0	205365.0	13.0	80.0	64.0	3.0

Distribution of data centre characteristics



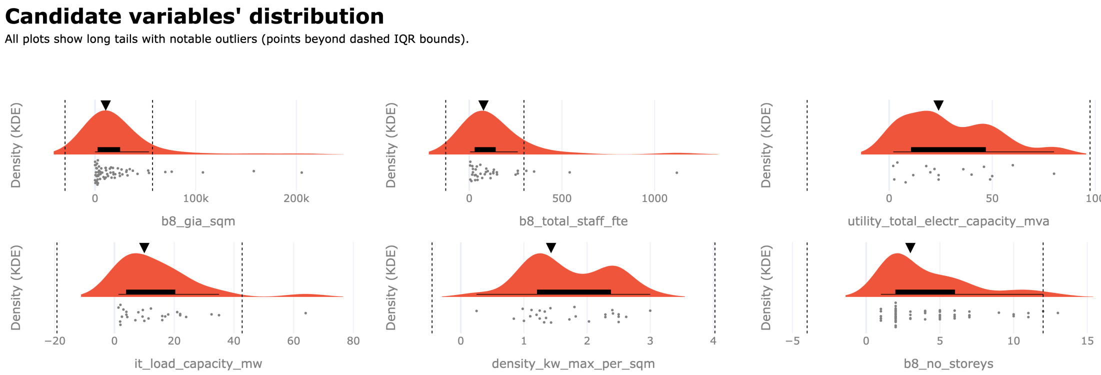


The dataset wide variation in the scale of London data centres. Median staff numbers are 77 FTE, compared with a mean of 137, while GIA ranges from 0.02 to 205.36 thousand sqm and site area from 0.01 to 13.03 ha. Technical capacity also varies substantially, with IT load ranging from 1.35 to 64 MW and electricity capacity from 2 to 80 MVA. The large gaps between medians and maximums indicate strong right-skew in several variables, while the differing counts highlight substantial missing data, particularly for the technical-capacity measures. More “full” columns need to be explored further.



The correlation matrix indicates strong positive relationships between several measures of data-centre scale, with many pairwise correlations ranging from approximately 0.69 to 0.85. Some relationships are weaker. An interesting exception is GIA per hectare, calculated as gross internal area divided by site area. Despite being derived directly from GIA and site area, this intensity measure shows relatively weak correlations with many of the other variables. This suggests that absolute facility size and development intensity capture different characteristics of data centres.

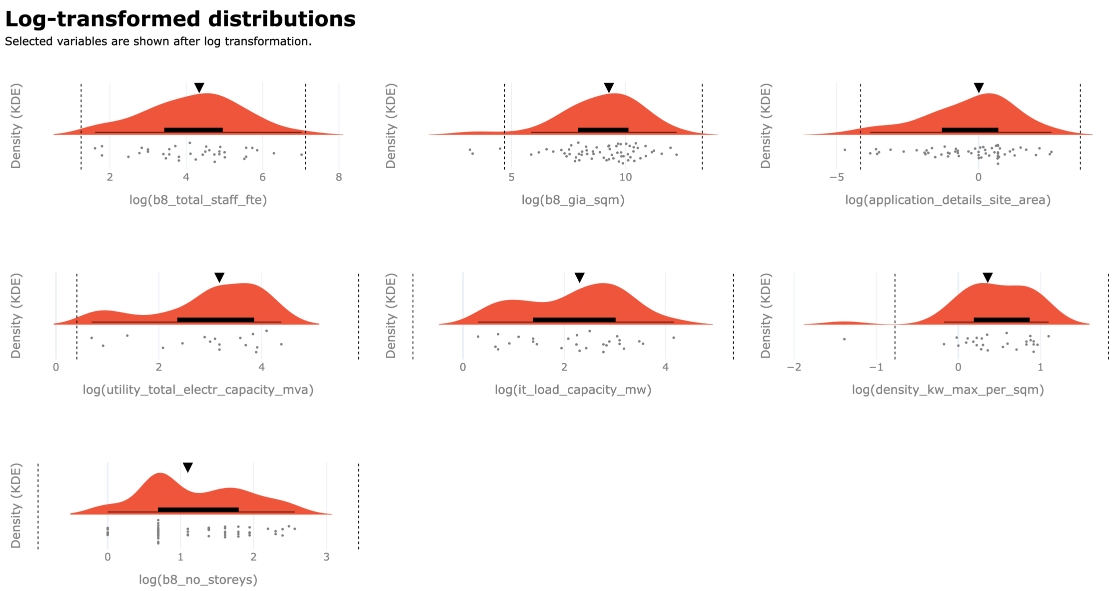
The stronger correlations between absolute scale measures, alongside the weaker relationships for the intensity measure, support further exploration of GIA as a predictor of staff numbers.



Relevant variables' distributions have long right tails with severe outliers.

Predictor screening

We begin by testing which characteristics of data centres are associated with staff numbers using separate OLS regression models. Starting hypothesis is that larger facilities, measured by GIA (b8_gia_sqm), require more staff. Each initial model uses the available complete observations for the variables being tested. Several variables are strongly right-skewed: log transformations are applied.




```
# ________________________________________________________________________
y_col = "b8_total_staff_fte"
x_cols = ["b8_gia_sqm", "application_details_site_area",
```

```
        "utility_total_electr_capacity_mva", "it_load_capacity_mw",
        "density_kw_max_per_sqm", "b8_no_storeys"]
results = []
for x_col in x_cols:
    value_cols = [y_col, x_col]
    d = data_ldn_exstg.dropna(subset=value_cols).copy()
    # log transformation
    d = d[(d[y_col] > 0) & (d[x_col] > 0)].copy()
    d["log_y"] = np.log(d[y_col])
    d["log_x"] = np.log(d[x_col])

    y = d["log_y"]
    X = sm.add_constant(d["log_x"])
    model = sm.OLS(y, X).fit()
    results.append({  # store model results
        "x_var": x_col,
        "n": int(model.nobs),
        "beta": model.params["log_x"],
        "p_value": model.pvalues["log_x"],
        "R2": model.rsquared,
        "Adj_R2": model.rsquared_adj,
        "intercept": model.params["const"],
    })
# ---------------------------------------------------------------------------
```

	x_var	n	beta	p_value	R2	Adj_R2	intercept
0	b8_gia_sqm	36	0.575	0.000	0.394	0.376	-1.29
1	b8_no_storeys	35	1.315	0.000	0.401	0.382	2.19
2	application_details_site_area	35	0.410	0.004	0.225	0.202	4.22
3	utility_total_electr_capacity_mva	10	0.691	0.058	0.380	0.302	1.76
4	it_load_capacity_mw	15	0.488	0.063	0.242	0.183	2.83
5	density_kw_max_per_sqm	15	0.178	0.825	0.004	-0.073	3.90

Across the separate log-log OLS models, GIA and number of storeys show the clearest positive relationships with staff numbers. A 1% increase in GIA is associated with about a 0.58% increase in staff numbers, while a 1% increase in the number of storeys is associated with about a 1.32% increase. Site area is also positively associated with staff numbers, but explains less variation.

Electricity capacity and IT load also have positive coefficients, but are not statistically significant at the 5% level and are based on much smaller samples.

We therefore retain GIA as a key explanatory variable because it is statistically significant, has a relatively large sample, and provides a direct measure of facility scale. However, the similarly strong result for storeys suggests that building configuration may matter alongside overall size. This supports testing GIA, site area and storeys together rather than assuming that staff numbers are driven by floorspace alone.

Baseline modelling

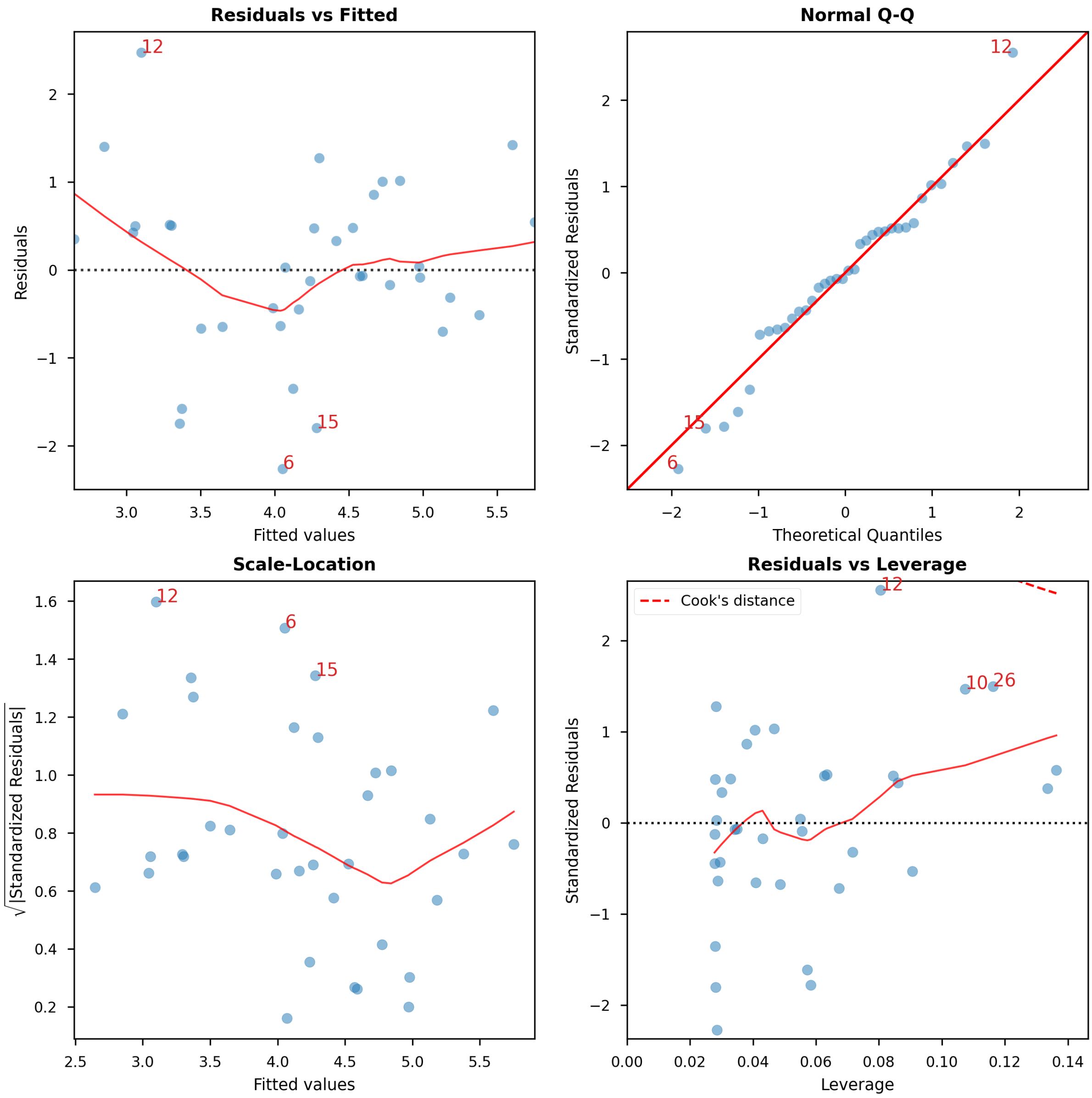
```
# ---------------------------------------------------------------------------
# Baseline Model: b8_total_staff_fte ~ b8_gia_sqm
```

```python
Xvar = "b8_gia_sqm"
yvar = "b8_total_staff_fte"

value_cols_m1 = [yvar, Xvar]
data_ldn_exstg_regression_m1 = data_ldn_exstg.dropna(subset=value_cols_m1).copy()

#taking the log
data_ldn_exstg_regression_m1 = data_ldn_exstg_regression_m1[
    (data_ldn_exstg_regression_m1[y_col] > 0) &
    (data_ldn_exstg_regression_m1[Xvar] > 0)].copy()
data_ldn_exstg_regression_m1["log_y"] = np.log(data_ldn_exstg_regression_m1[yvar])
data_ldn_exstg_regression_m1["log_x"] = np.log(data_ldn_exstg_regression_m1[Xvar])
final_y = "log_y"
final_x = "log_x"
# ---------------------------------------------------------------------------
```

After removing these incomplete observations, the sample was reduced from 67 to 36 observations.

```python
# ---------------------------------------------------------------------------
y = data_ldn_exstg_regression_m1[final_y]
X = sm.add_constant(data_ldn_exstg_regression_m1[final_x])

m1 = sm.OLS(y, X).fit()
m1_robust_se = sm.OLS(y, X).fit(cov_type="HC3")
#a model variation with heteroskedasticity-robust standard errors
m1_robust_HuberT = sm.RLM(y, X, M=sm.robust.norms.HuberT()).fit()
#a model variation that downweights influential outliers (see diagnostics below)).
#We confirmed these are not errors.
# ---------------------------------------------------------------------------
```

	Model	N	Intercept	GIA coef.	Std. error	p-value	R^2
0	OLS	36	-1.29	0.575473	0.12	0.0	0.394
1	OLS (HC3)	36	-1.29	0.575473	0.13	0.0	0.394
2	Robust (HuberT)	36	-1.30	0.581148	0.11	0.0	NaN

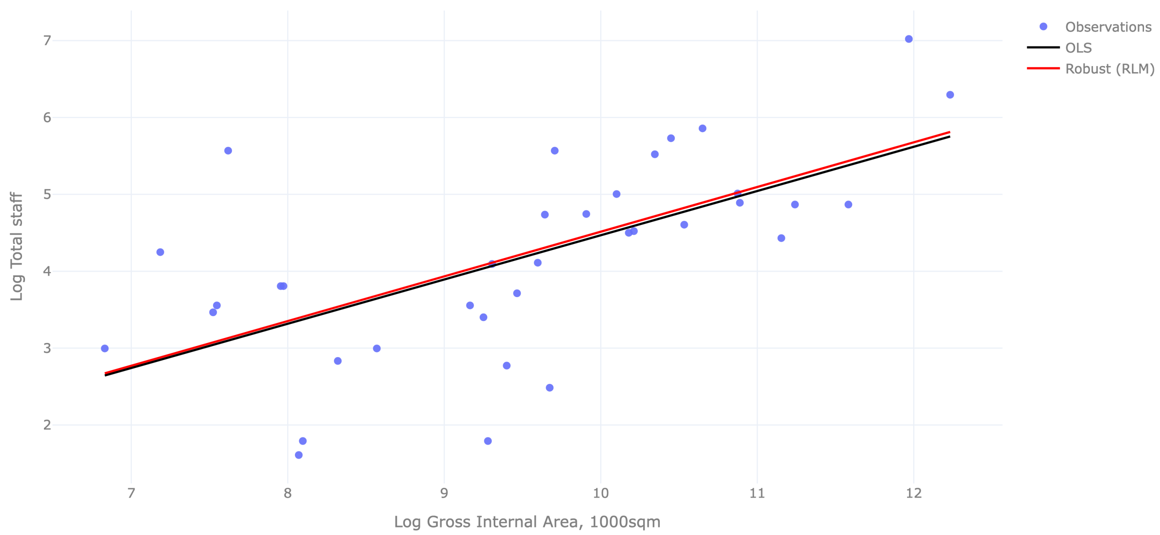


	Features	VIF Factor
1	log_x	1.00
0	const	48.85

Regression diagnostics were used to assess the log-log OLS specification. HC3 robust standard errors were used to provide inference that is less sensitive to heteroskedasticity, while Huber robust regression was estimated as a sensitivity check for influential observations. The estimated coefficient is highly consistent across OLS, HC3 and Huber specifications (approximately 0.58), suggesting that the positive relationship is not driven by a small number of influential observations.

A **1% increase in GIA is associated with approximately a 0.58% increase in staff numbers. GIA alone accounts for around 39% of the variation in log staff numbers** in the analysed sample. As the coefficient is below 1, the relationship is sub-proportional: staff numbers increase with facility size, but at a slower rate.

Baseline regression models' fitted lines



Mapped residuals, baseline model staff numbers - gross internal area

Blue points: negative residuals, white: near zero, red: positive residuals.

Mapped residuals suggest some spatial patterning: positive residuals appear around visibly clustered locations such as Canary Wharf and Southall, while other areas show a mix of positive and negative residuals.

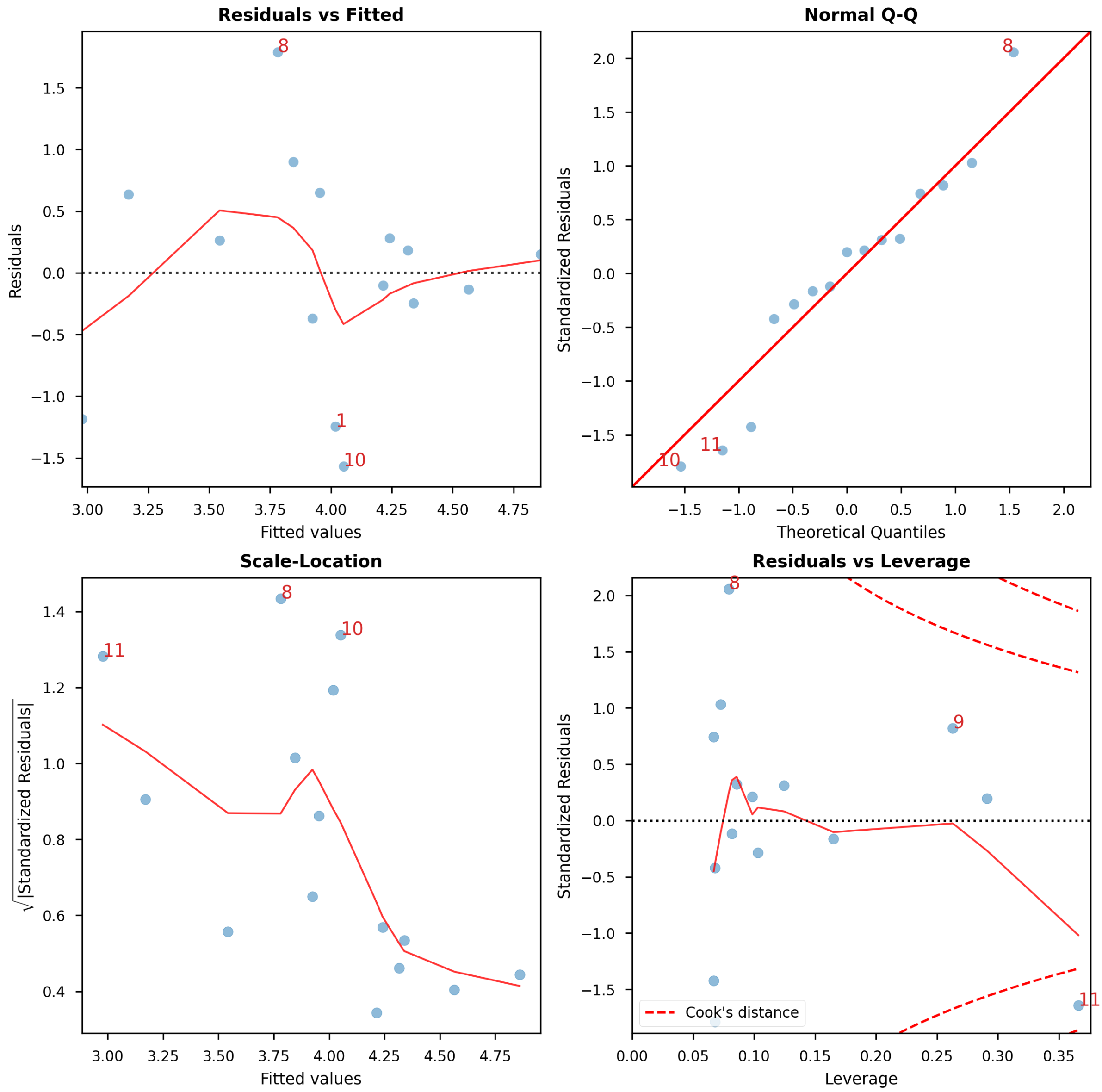
Alternative explanatory variable

We also test IT load capacity as an alternative explanatory variable because, among the non-area measures, it showed one of the closest p-values to the 5% significance threshold in the initial screening.

Compared with GIA, IT load shows a weaker and less stable relationship with staff numbers. Its coefficient is still positive (=0.488), but the model explains less variation (R^2=0.242 versus 0.394 for GIA) and the association is not statistically significant under OLS or HC3. The Huber model reaches significance, suggesting that the result may be sensitive to influential observations. We conclude that GIA provides stronger and more consistent evidence of an association with staff numbers, while the IT load result remains suggestive but uncertain, particularly given the much smaller sample (n=15 versus n=36).

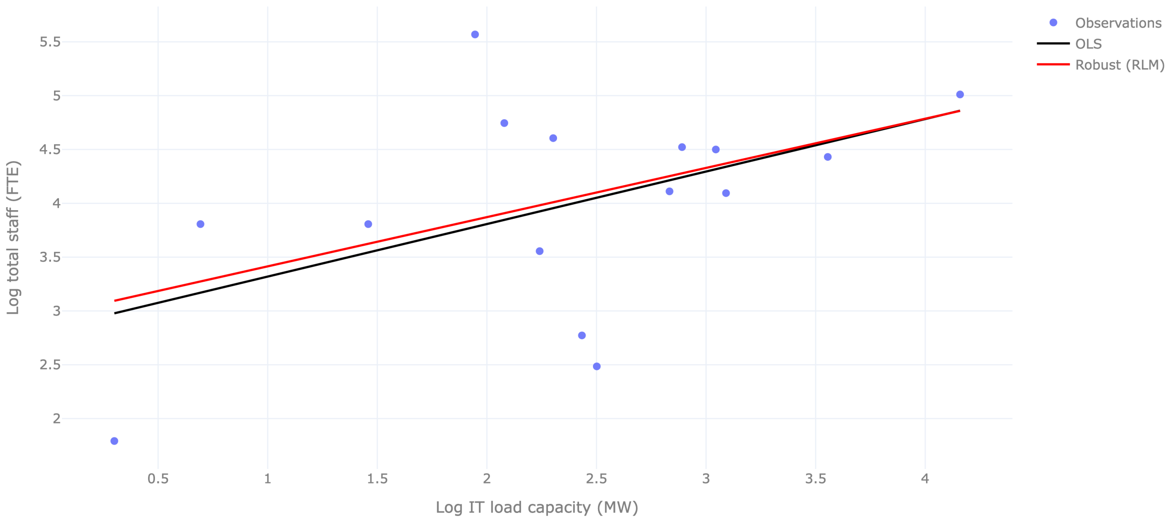
```
# ---------------------------------------------------------------------------
# Log-log regression:log(staff) ~ log(IT load)
yvar_m3 = "b8_total_staff_fte"
xvar_m3 = "it_load_capacity_mw"

data_ldn_exstg_regression_m3 = (
    data_ldn_exstg
    .dropna(subset=[yvar_m3, xvar_m3])
    .copy()
)

# log requires positive values
data_ldn_exstg_regression_m3 = data_ldn_exstg_regression_m3[
    (data_ldn_exstg_regression_m3[yvar_m3] > 0) &
    (data_ldn_exstg_regression_m3[xvar_m3] > 0)
].copy()

data_ldn_exstg_regression_m3["log_staff_m3"] = np.log(
    data_ldn_exstg_regression_m3[yvar_m3]
)
data_ldn_exstg_regression_m3["log_itload_m3"] = np.log(
    data_ldn_exstg_regression_m3[xvar_m3]
)

y_m3 = data_ldn_exstg_regression_m3["log_staff_m3"]
X_m3 = sm.add_constant(data_ldn_exstg_regression_m3[["log_itload_m3"]])

m3 = sm.OLS(y_m3, X_m3).fit()
m3_robust_se = sm.OLS(y_m3, X_m3).fit(cov_type="HC3")
m3_robust_HuberT = sm.RLM(
    y_m3, X_m3, M=sm.robust.norms.HuberT()
).fit()

models_m3 = [m3, m3_robust_se, m3_robust_HuberT]
# ---------------------------------------------------------------------------
```

	Model	N	Intercept	IT load coef.	IT load SE	IT load p	R^2
0	OLS	15	2.832	0.488	0.240	0.063	0.242
1	OLS (HC3)	15	2.832	0.488	0.299	0.102	0.242
2	Robust (HuberT)	15	2.957	0.457	0.227	0.044	NaN



	Features	VIF Factor
1	log_itload_m3	1.00
0	const	6.89

IT load and staff numbers: fitted regression lines



Further examination of predictors

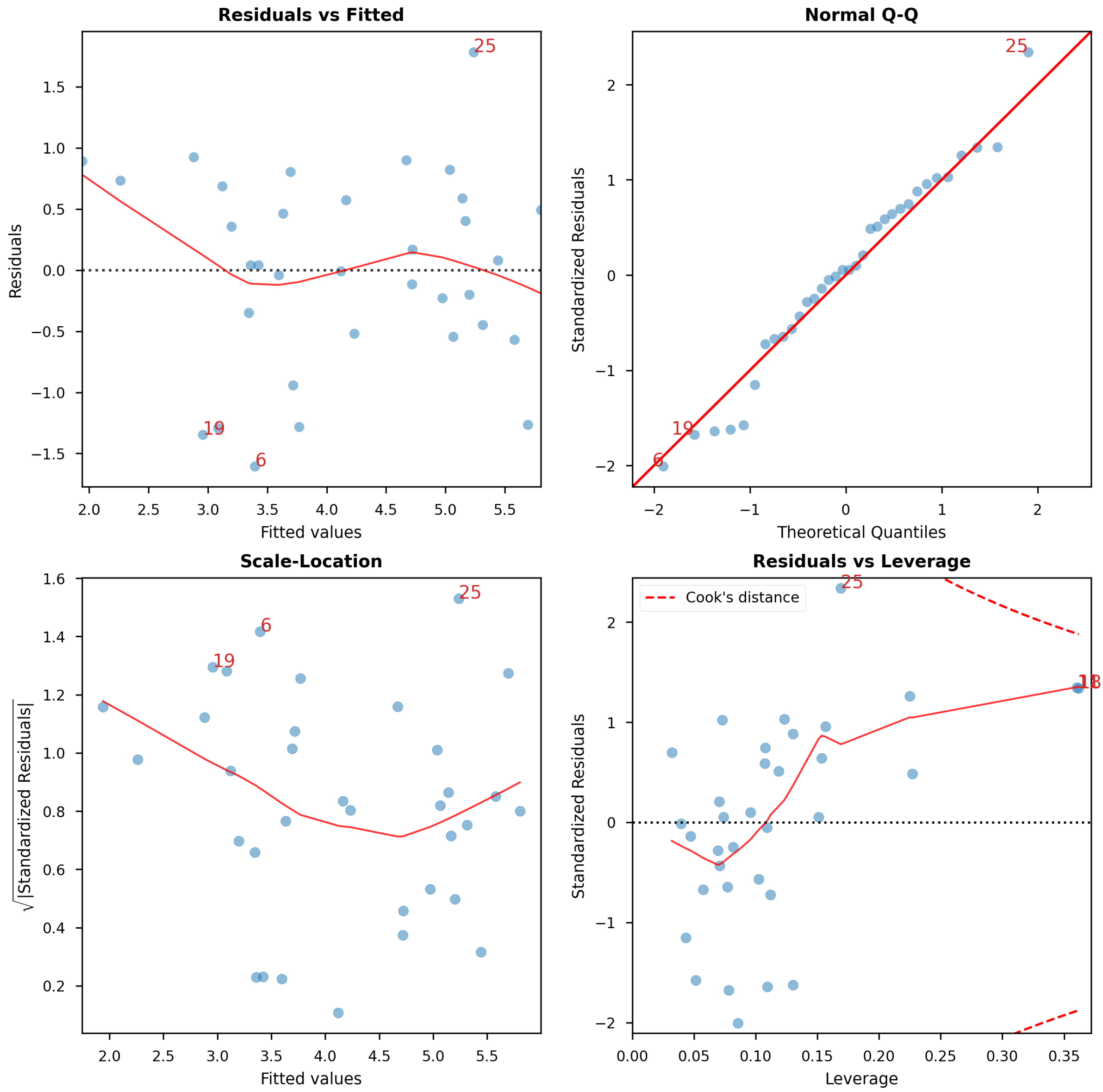
Because missing data substantially reduces the sample size, only variables with relatively complete coverage can be retained for this step. We therefore test whether the relationship between GIA and staff numbers changes when site area and number of storeys are included in the same model.

```
# ---------------------------------------------------------------------------
# Multiple log-log regression:log(staff) ~ log(site area) + log(GIA) + log(storeys)

Xvars_m2 = ["application_details_site_area", "b8_gia_sqm", "b8_no_storeys"]
yvar_m2 = "b8_total_staff_fte"
data_ldn_exstg_regression_m2 = (
    data_ldn_exstg
    .dropna(subset=[yvar_m2] + Xvars_m2)
    .copy()
)

# log transformation requires positive values
data_ldn_exstg_regression_m2 = data_ldn_exstg_regression_m2[
    (data_ldn_exstg_regression_m2[[yvar_m2] + Xvars_m2] > 0).all(axis=1)
].copy()

data_ldn_exstg_regression_m2["log_staff_m2"] = np.log(
    data_ldn_exstg_regression_m2[yvar_m2])
data_ldn_exstg_regression_m2["log_site_area_m2"] = np.log(
    data_ldn_exstg_regression_m2["application_details_site_area"])
data_ldn_exstg_regression_m2["log_gia_m2"] = np.log(
    data_ldn_exstg_regression_m2["b8_gia_sqm"])
data_ldn_exstg_regression_m2["log_storeys_m2"] = np.log(
    data_ldn_exstg_regression_m2["b8_no_storeys"])
```

```python
y_m2 = data_ldn_exstg_regression_m2["log_staff_m2"]
X_m2 = sm.add_constant(
    data_ldn_exstg_regression_m2[
        ["log_site_area_m2", "log_gia_m2", "log_storeys_m2"]
    ]
)

m2 = sm.OLS(y_m2, X_m2).fit()
m2_robust_se = sm.OLS(y_m2, X_m2).fit(cov_type="HC3")
m2_robust_HuberT = sm.RLM(y_m2, X_m2, M=sm.robust.norms.HuberT()).fit()
# -------------------------------------------------------------------------
```

Table 1

Term	OLS	OLS (HC3)	Robust (HuberT)
Intercept	1.141	1.141	1.546
Site area	0.327 (SE 0.148, p=0.035)	0.327 (SE 0.194, p=0.093)	0.321 (SE 0.156, p=0.039)
GIA	0.134 (SE 0.188, p=0.481)	0.134 (SE 0.256, p=0.599)	0.096 (SE 0.198, p=0.627)
Storeys	1.192 (SE 0.296, p=0.000)	1.192 (SE 0.402, p=0.003)	1.177 (SE 0.312, p=0.000)
R^2	0.631	0.631	—



	Features	VIF Factor
3	log_storeys_m2	1.64
1	log_site_area_m2	2.39
2	log_gia_m2	3.02
0	const	120.75

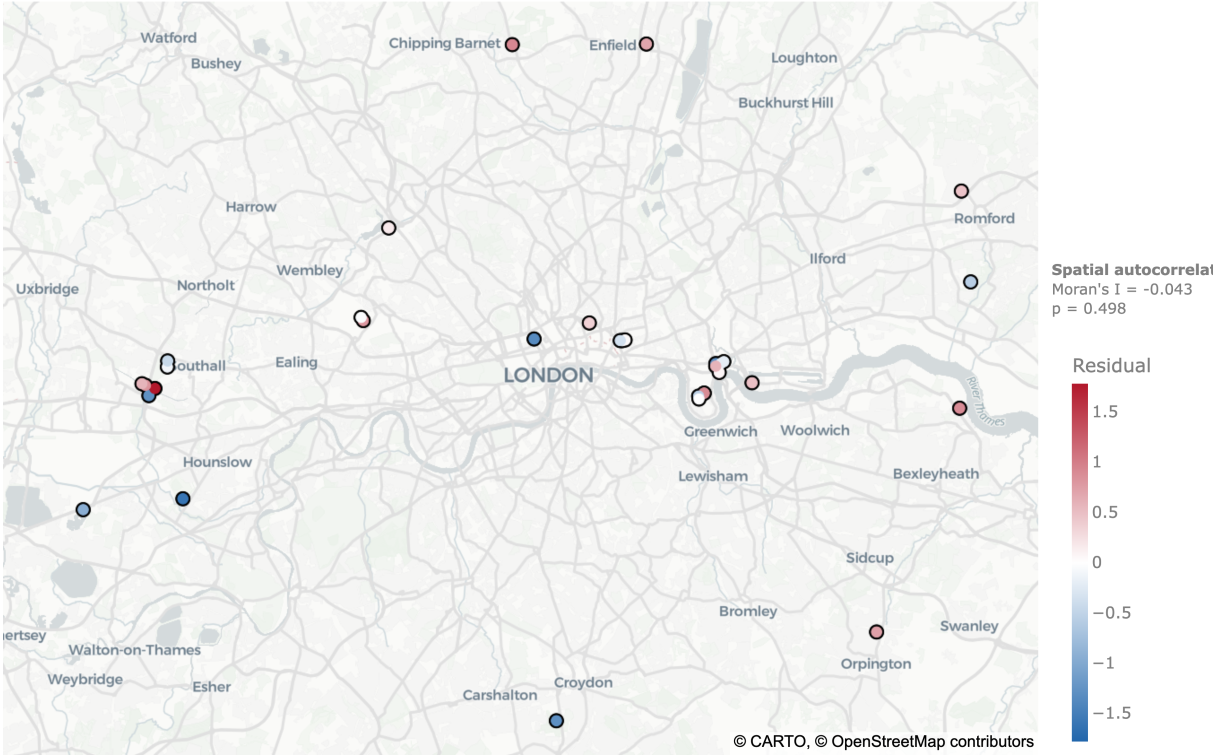
Compared with the GIA-only model, adding site area and number of storeys increases R^2 substantially, from about 0.39 to 0.63. GIA itself becomes small and non-significant, while number of storeys emerges as the strongest predictor; site area remains positive but is less robust under HC3.

Multicollinearity remains moderate with VIF values between 1.64 and 3.02. Even so, around 37% of the variation in log staff numbers remains unexplained, indicating that factors beyond physical size and built form are likely to matter.

The Q–Q plot shows some departure from normality in the residual tails, although most observations follow the expected distribution reasonably closely.

Mapped residuals, extended model

Includes GIA, site area and number of storeys.



The extended model residuals show no significant spatial autocorrelation (Moran's I = -0.043, $p = 0.498$). Differences in data centre operating models are not captured in the dataset and therefore cannot be tested.